PTP and gPTP

Precision Time Protocol (IEEE 1588) and

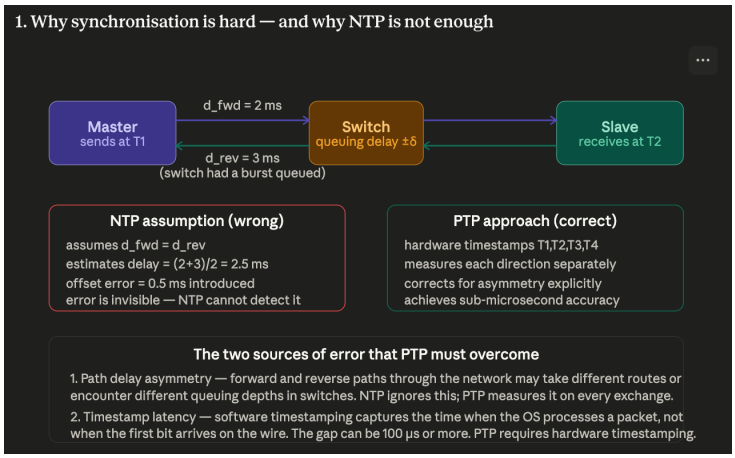
Generalised PTP for Time-Sensitive Networks (IEEE 802.1AS)

A Complete Technical Reference

Covering: synchronisation fundamentals · four-timestamp mathematics · hardware timestamping · BMCA · clock types · gPTP peer delay · correctionField propagation · servo control · automotive deployment

1. Why Synchronisation Is Hard — and Why NTP Is Not Enough

Network Time Protocol (NTP) achieves millisecond-level accuracy over the internet by averaging many round-trip measurements and assuming that the forward and reverse path delays are equal (symmetric). These two assumptions break down when microsecond accuracy is needed.



The asymmetric delay problem

Consider two nodes exchanging timing messages through a switch. The forward path may take 2 ms and the reverse path 3 ms — because the switch had a burst of queued traffic in one direction but not the other. NTP calculates the one-way delay as (2 + 3) / 2 = 2.5 ms and applies a 2.5 ms correction. The true delay in the forward direction was only 2 ms, so the NTP-corrected slave clock is 0.5 ms wrong. Critically, NTP cannot detect this error — it has no way to measure each direction independently.

> NTP error: assumes forward delay = reverse delay. If the switch queuing is asymmetric, the error is silent and uncorrectable from the slave's perspective.

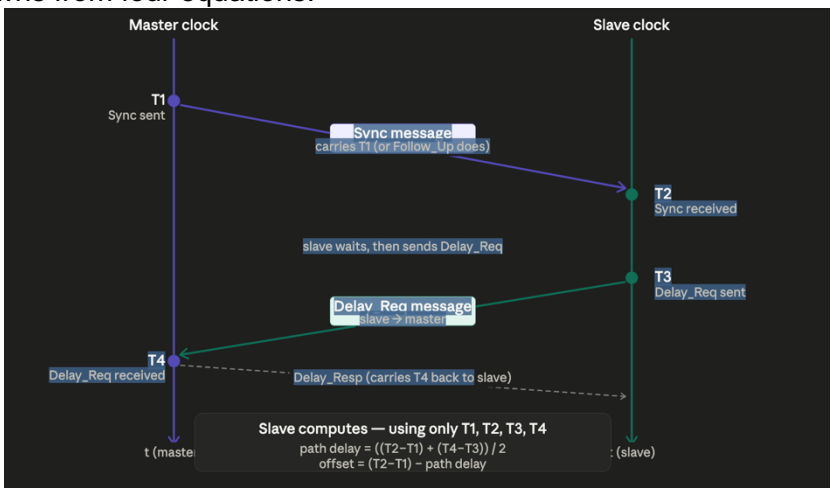
The software timestamp problem

NTP and naive PTP implementations record the time when the operating system (OS) processes a received packet — not when the first bit arrived at the network interface. The gap between actual packet arrival and OS processing can be 50 to 500 microseconds, and it varies unpredictably with CPU load, interrupt latency, and cache state. This alone limits any software-timestamped protocol to millisecond accuracy at best.

PTP (Precision Time Protocol, IEEE 1588) solves both problems: it measures each direction of the path delay separately using four hardware timestamps, and it captures those timestamps at the physical layer boundary (the MDI — Media Dependent Interface) the instant the Start Frame Delimiter (SFD) byte crosses the wire.

2. The Four Timestamps — The Mathematical Foundation of PTP

Every PTP synchronisation exchange produces exactly four hardware timestamps. These four numbers are sufficient to simultaneously solve for both the clock offset and the one-way path delay — two unknowns from four equations.



The timestamp exchange sequence

Timestamp	Who captures it	What it records
T1	Master PHY hardware	Exact moment the master transmits the Sync message (SFD crosses the master MDI outbound)
T2	Slave PHY hardware	Exact moment the slave receives the Sync message (SFD crosses the slave MDI inbound)
T3	Slave PHY hardware	Exact moment the slave transmits the Delay_Req message (SFD crosses the slave MDI outbound)
T4	Master PHY hardware	Exact moment the master receives the Delay_Req message (SFD crosses the master MDI inbound)

The mathematics

From these four timestamps, the slave computes two quantities:

```
path_delay  =  ((T2 - T1) + (T4 - T3)) / 2
clock_offset  =  (T2 - T1) - path_delay
```

The elegance of this formulation is that the path delay cancels out of the offset calculation, and the clock offset cancels out of the path delay calculation. The two unknowns are fully determined by the four measurements regardless of the magnitude of either.

Worked example with real numbers

Suppose the master clock is the ground truth. The slave clock is running 10 microseconds (µs) ahead of the master (offset = +10 µs), and the one-way propagation delay is exactly 5 µs in each direction.

Timestamp	Value (µs)	Derivation
T1	1000.000	Master sends Sync at this master-clock time
T2	1015.000	Slave receives it: 1000 + 5 (delay) + 10 (offset) = 1015 on slave clock
T3	2000.000	Slave sends Delay_Req at this slave-clock time
T4	1995.000	Master receives it: 2000 - 10 (offset) + 5 (delay) = 1995 on master clock

Applying the formulae:

```
path_delay  =  ((1015 - 1000) + (1995 - 2000)) / 2  =  (15 + (-5)) / 2  =  5 µs  ✓

        clock_offset  =  (1015 - 1000) - 5  =  15 - 5  =  +10 µs  ✓
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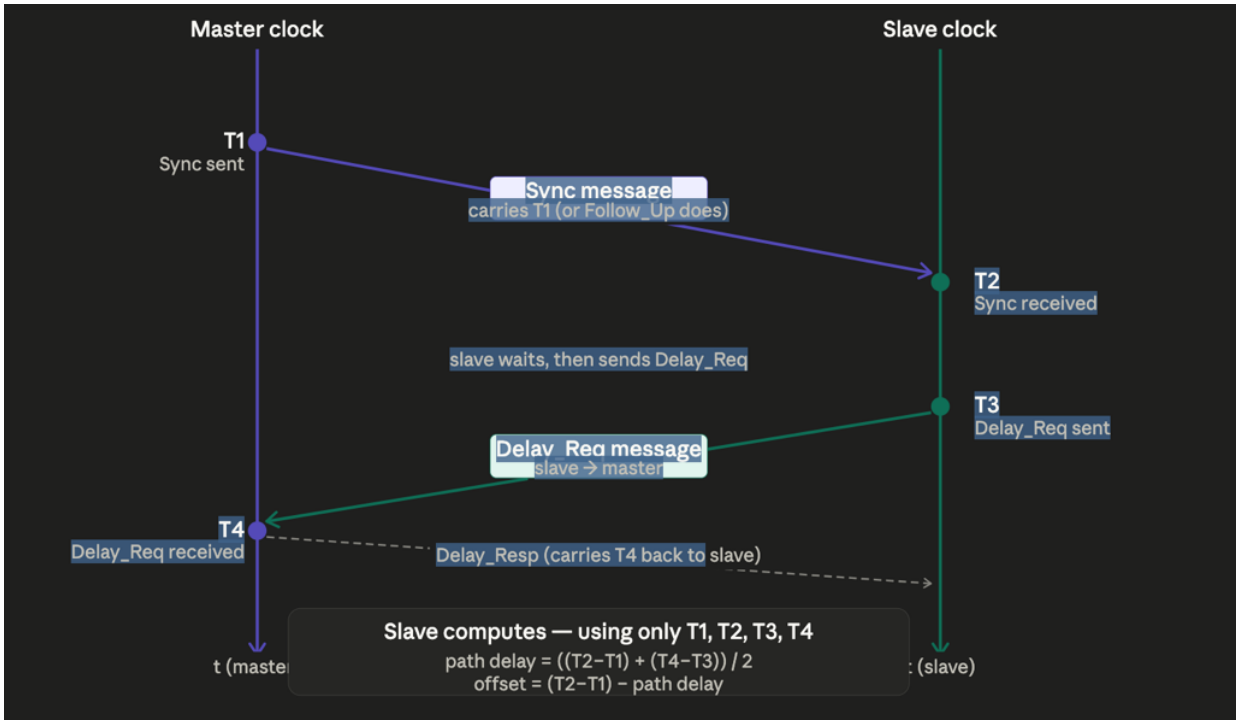
The slave now knows it is 10 µs ahead and corrects its clock accordingly. Both results are exact — the arithmetic is not an approximation.

3. Hardware Timestamping ---*Why Software Timestamps Fail*



The hardware timestamp unit is a small register in the PHY or MAC silicon that latches the current value of a free-running nanosecond counter the instant the SFD is detected at the MDI. This happens entirely independently of the CPU, OS scheduler, interrupt handler, and application software.

Approach	Where timestamp is captured	Typical accuracy
Software timestamping	Inside the OS network stack, after interrupt processing, driver execution, and kernel scheduling	50–500 µs — variable and unpredictable
NIC-level timestamping	In the NIC driver at the point of DMA completion — better than pure software but still above the MAC	1–10 µs — improved but insufficient for PTP
Hardware (PHY/MAC) timestamping	At the MDI boundary, the instant the SFD crosses the wire interface — in silicon, no software involvement	< 8 ns — sufficient for sub-microsecond PTP accuracy

> IEEE 802.1AS (gPTP) makes hardware timestamping mandatory at every port. Without it, gPTP compliance cannot be achieved. This is also why gPTP-capable switch ASICs contain a dedicated PTP timestamp unit — separate from the normal forwarding pipeline — that captures SFD arrival times to nanosecond resolution.

How the hardware timestamp unit works

The PHY runs a free-running 64-bit counter clocked from its recovered line clock (for receive) or its local crystal reference (for transmit). When the SFD pattern (0xAB in an Ethernet frame) is detected, the counter value is latched into a dedicated register. The PTP software reads this register via the MDIO (Management Data Input/Output) interface after transmission or reception. The counter frequency is typically 125 MHz (8 ns resolution) or higher.

4. PTP Message Types — The Complete Protocol Vocabulary

PTP defines two categories of message: event messages (which are hardware-timestamped on both transmit and receive) and general messages (which carry information but do not require timestamping).

Event messages — hardware timestamped

Message	Direction	Purpose
Sync	Master → Slave	Primary timing message. Carries T1 (the master's send timestamp) either embedded in the message (one-step mode) or via a subsequent Follow_Up message (two-step mode). The slave captures T2 on receipt.
Delay_Req	Slave → Master	Initiates the reverse path delay measurement. The slave captures T3 on transmission. Sent at a lower rate than Sync — typically once per second.
Pdelay_Req	Port → Neighbour	Used in the peer-to-peer delay mechanism (mandatory in gPTP). Initiates a link delay measurement between directly connected ports. Captures T1 on transmission.

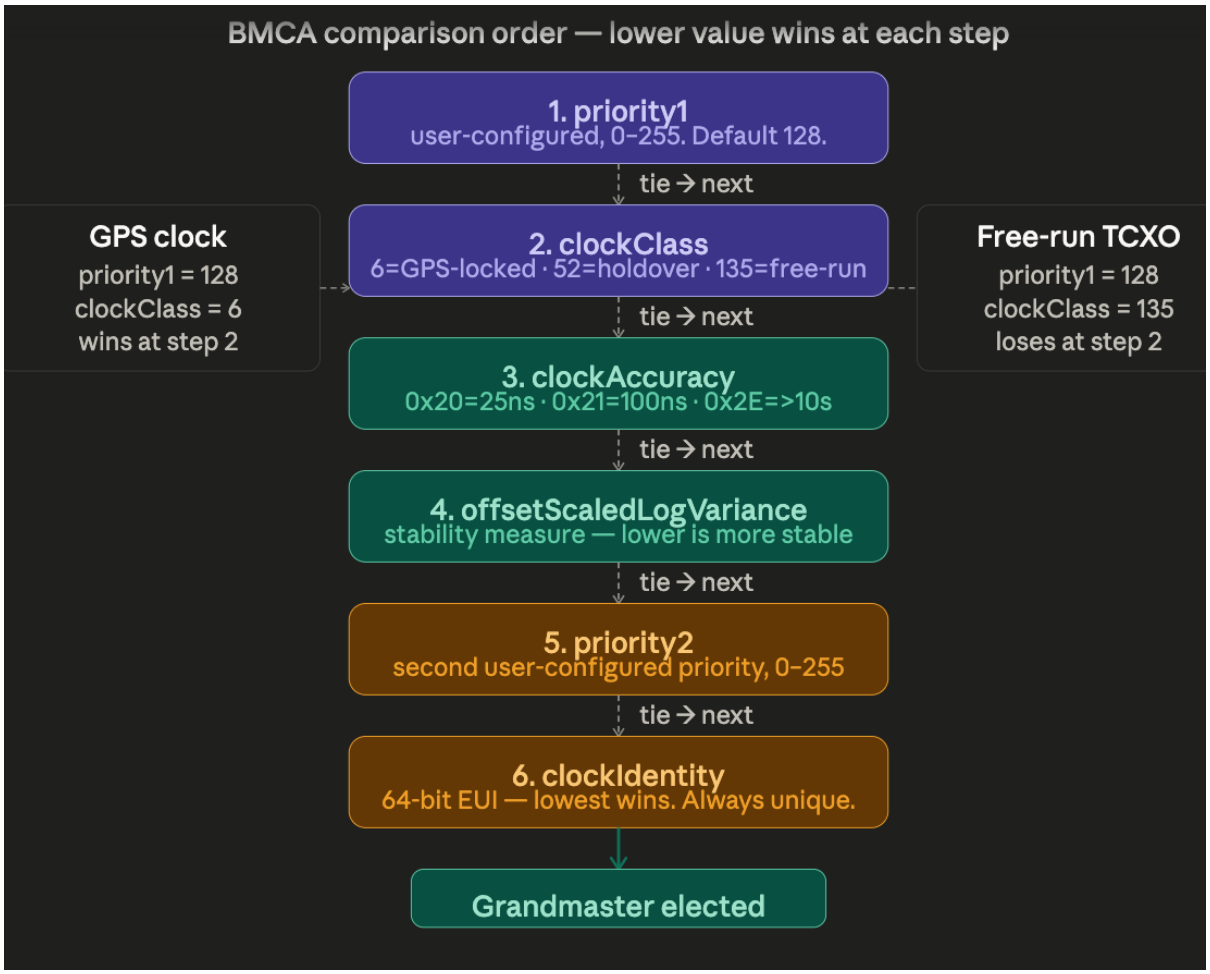
Message	Direction	Purpose
Pdelay_Resp	Neighbour → Port	Response to Pdelay_Req. The responding port captures T2 (receive time of Pdelay_Req) and T3 (send time of Pdelay_Resp). Both are hardware timestamped.

General messages — not timestamped

Message	Direction	Purpose
Follow_Up	Master → Slave	Carries the precise T1 hardware timestamp for the preceding Sync when two-step clock mode is used. The master reads the hardware timestamp register after transmission and sends it in this message.
Delay_Resp	Master → Slave	Carries T4 (master's receive timestamp of the Delay_Req) back to the slave. After receiving this, the slave has all four timestamps.
Pdelay_Resp_Follow_Up	Neighbour → Port	Carries the precise T3 hardware timestamp for the Pdelay_Resp in two-step mode. Completes the peer delay exchange.
Announce	Master → All	Carries the grandmaster's clock quality attributes used by the Best Master Clock Algorithm (BMCA) to elect the grandmaster. Every potential master sends Announce periodically.
Management	Bidirectional	Read and write PTP configuration and state variables on remote clocks. Optional in IEEE 1588-2019.
Signaling	Bidirectional	Negotiate unicast transmission contracts — a slave requests that a master send Sync at a specific rate. Used in large-scale unicast PTP deployments.

5. The Best Master Clock Algorithm (BMCA) — Electing the Grandmaster

The BMCA (Best Master Clock Algorithm) is the election protocol that determines which clock becomes the grandmaster. It runs locally and independently on every PTP-capable device — there is no centralised election server. Every device that receives Announce messages applies the same deterministic comparison and converges to the same answer.



Comparison order — lower value wins at each step

Two candidate clocks are compared in strict priority order. The first attribute that differs determines the winner. If all attributes are equal, the comparison continues to the next level.

Step	Attribute	Range / typical values	Notes
1	priority1	0–255, default 128	User-configured. The single most powerful override — set priority1=1 to force a clock to always win regardless of quality.
2	clockClass	6 = GPS-locked; 52 = holdover; 135 = free-running oscillator	The most important attribute in practice. A GPS-disciplined clock (class 6) always beats a free-running oscillator (class 135) regardless of priority2.
3	clockAccuracy	0x20 = 25 ns; 0x21 = 100 ns; 0x2E = > 10 s	Encoded enumeration. Reports the clock's accuracy relative to UTC.
4	offsetScaledLogVariance	Lower = more stable	A measure of the clock's frequency stability (Allan deviation). More stable clocks win.

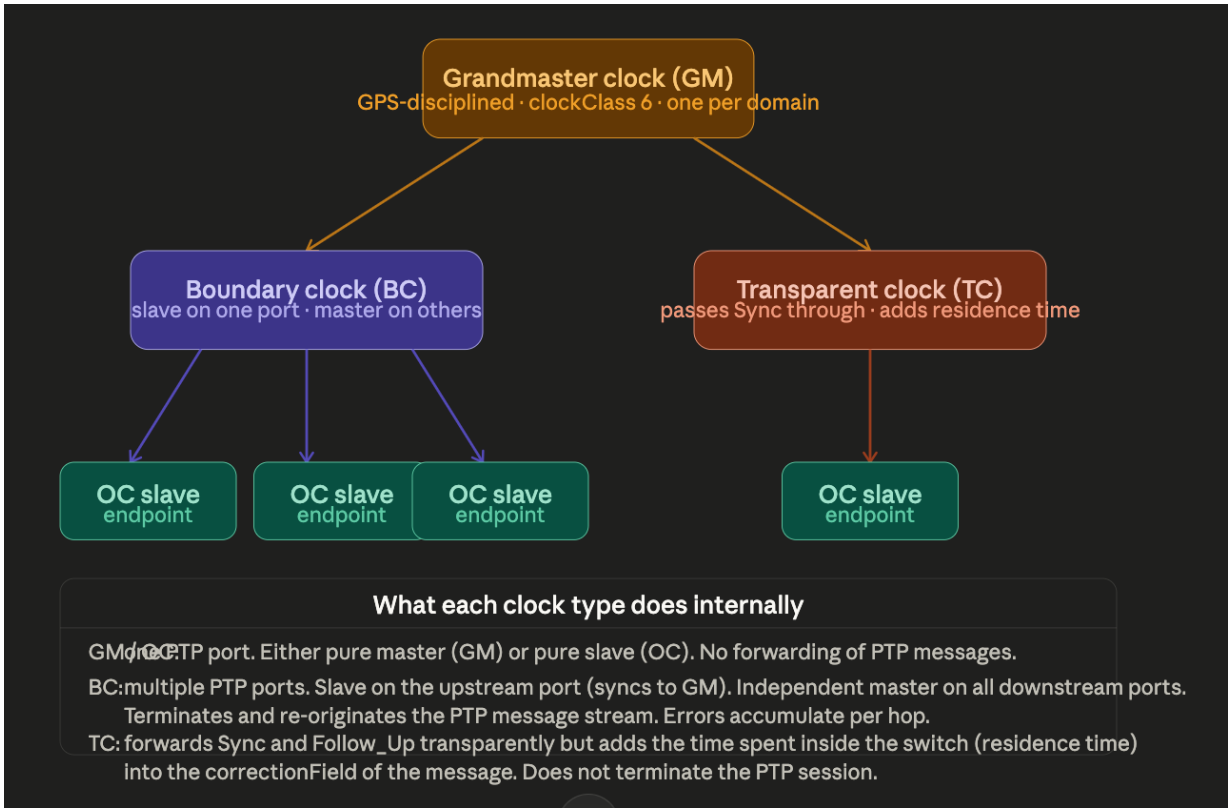
Step	Attribute	Range / typical values	Notes
5	priority2	0–255, default 128	Second user-configured priority. Used to break ties between clocks of the same quality class.
6	clockIdentity	64-bit EUI (Extended Unique Identifier)	Derived from the device MAC address. Always unique — guarantees a deterministic result even if all other attributes are identical.

> Practical rule: any clock with clockClass 6 (GPS-locked) will automatically become grandmaster on any network it joins, provided its priority1 is not set higher than 127. No manual configuration is needed — the BMCA handles it.

What happens when the grandmaster fails

If the grandmaster stops sending Announce messages, all devices start an Announce receipt timeout timer (typically 3 × announce interval). When the timer expires, every device runs the BMCA again using the remaining Announce messages. The next-best clock becomes the new grandmaster within seconds. No manual failover is required.

6. PTP Clock Types — Grandmaster, Ordinary, Boundary, and Transparent



PTP defines four clock types, each with a distinct role in distributing time through a network. Understanding them is essential for designing a PTP deployment correctly.

Clock type	Acronym	Ports	Role
Grandmaster clock	GM	One (master only)	The network's single authoritative time source. Typically disciplined by a GPS/GNSS receiver, which gives it clockClass 6 and traceable accuracy to UTC. Sends Sync messages to all slaves. There is exactly one active grandmaster per PTP domain.
Ordinary clock	OC	One (master or slave)	The simplest clock type — one PTP port, either pure master or pure slave. Does not forward or relay PTP messages. Every endpoint ECU in a vehicle (camera, radar, brake controller) is an OC operating in slave mode. Receives Sync messages and runs the servo to align its local clock.
Boundary clock	BC	Multiple	Slave on its upstream port (synchronises to the grandmaster), independent master on all downstream ports. Terminates and re-originates the PTP session at each hop. Downstream slaves see the boundary clock as their master, not the grandmaster. Errors accumulate per hop but the topology scales to large networks.
Transparent clock	TC	Multiple (switch)	Passes Sync messages through without terminating the PTP session. Measures its own frame residence time (how long the frame spent inside the switch) and adds it to the correctionField of the Sync message. The downstream slave can then subtract the accumulated correctionField and effectively see the grandmaster directly. More accurate than a boundary clock — errors do not accumulate.

The correctionField — how transparent clocks account for themselves

When a Sync message enters a transparent clock switch, the hardware records its ingress timestamp. When the Sync leaves on the egress port, the hardware records the egress timestamp. The difference — the residence time — is added to the correctionField in the frame before it is forwarded. The downstream slave subtracts correctionField from its receive timestamp and recovers the grandmaster's original send time, as if the switch had zero latency.

> Example: Sync enters switch at 1000.000 µs, exits at 1000.800 µs. Residence time = 800 ns. Switch adds 800 ns to correctionField. Slave receives frame, subtracts 800 ns from its calculation. From the slave's perspective, the switch is transparent.

7. gPTP (IEEE 802.1AS) — The TSN Profile of PTP

gPTP (Generalised Precision Time Protocol) is not a separate protocol — it is a precisely constrained profile of IEEE 1588, specifically designed for bridged IEEE 802 LAN environments including automotive networks, industrial automation, and professional audio/video. It makes mandatory several choices that IEEE 1588 leaves optional, and adds the concept of per-port link delay measurement.

Key differences: gPTP versus IEEE 1588

Aspect	IEEE 1588-2019	IEEE 802.1AS-2025 (gPTP)
Network scope	Any IP or Ethernet network, including WAN	Bridged IEEE 802 LANs only — single network segment
Transport	UDP/IPv4, UDP/IPv6, or Ethernet Layer 2	Ethernet Layer 2 only (EtherType 0x88F7) — no IP overhead
Delay mechanism	Either end-to-end (E2E) or peer-to-peer (P2P) — operator chooses	Peer-to-peer (P2P) only — mandatory. Each port measures its direct neighbour independently.
Hardware timestamps	Recommended but not mandatory	Mandatory at every port. Non-negotiable for compliance.
One-step vs two-step	Both permitted	Both permitted; one-step strongly preferred to reduce message count
Clock model	GM + BC + TC + OC	Every bridge port is a time-aware system. All bridges act as TC with per-port link delay correction.
Accuracy target	Sub-microsecond to tens of nanoseconds (hardware dependent)	± 1 µs required across the full vehicle network (up to 8 hops)
Typical domain	Up to 128 independent clock domains	Domain 0 only in automotive — all nodes share one domain

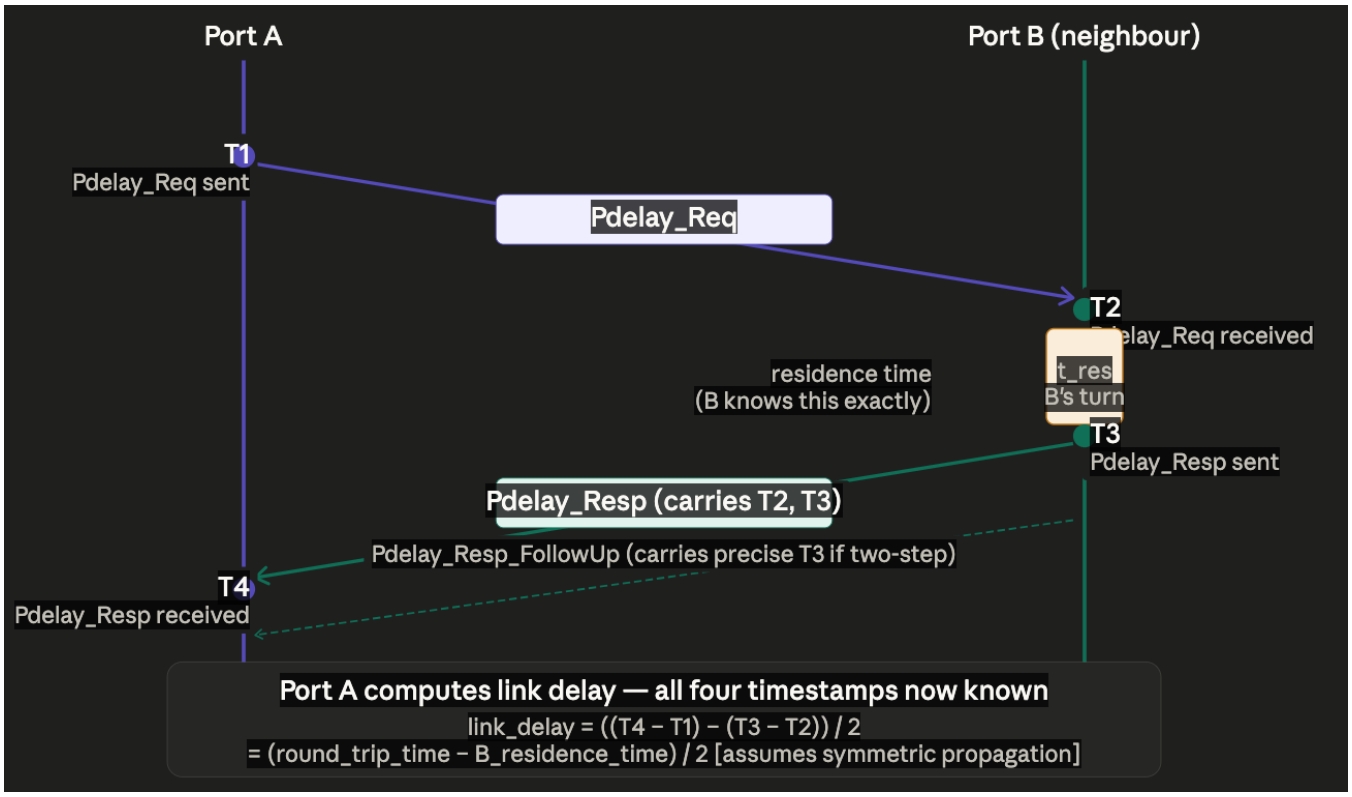
Why peer-to-peer delay is mandatory in gPTP

In the end-to-end delay mechanism (used by IEEE 1588), the slave sends a Delay_Req all the way to the grandmaster and waits for a response. In a large network with many hops, this creates significant traffic and the path delay measurement reflects the entire network path, not individual links.

In the peer-to-peer mechanism, every port measures the delay to its direct neighbour independently and continuously. This has three advantages for TSN networks: it is faster to converge (only one link hop per measurement), it is more accurate (no accumulated error from intermediate switches), and it gives each switch precise knowledge of its own link delays — essential for computing correctionField values accurately when forwarding Sync messages.

8. gPTP Peer Delay Measurement — The Per-Link Engine

The peer delay mechanism is the continuous, per-port measurement engine that underlies all of gPTP's accuracy. Every gPTP port measures the propagation delay to its direct neighbour using a four-timestamp exchange analogous to the main Sync/Delay_Req cycle, but running independently on each link.



Timestamp	Who captures	What it records
T1	Port A hardware	Port A sends Pdelay_Req — SFD crosses Port A's MDI outbound
T2	Port B hardware	Port B receives Pdelay_Req — SFD crosses Port B's MDI inbound
T3	Port B hardware	Port B sends Pdelay_Resp — SFD crosses Port B's MDI outbound
T4	Port A hardware	Port A receives Pdelay_Resp — SFD crosses Port A's MDI inbound

The link delay formula

```
link_delay = ((T4 - T1) - (T3 - T2)) / 2
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This is: (total round-trip time minus Port B's residence time) divided by 2. Port B's residence time (T3 - T2) is known exactly because Port B measured both T2 and T3 with its own hardware. The division by 2 assumes symmetric propagation — that the signal takes the same time in each direction. This assumption is valid for point-to-point links but can introduce small errors on links with asymmetric media (e.g. fibre with different transmit and receive wavelengths).

Worked example

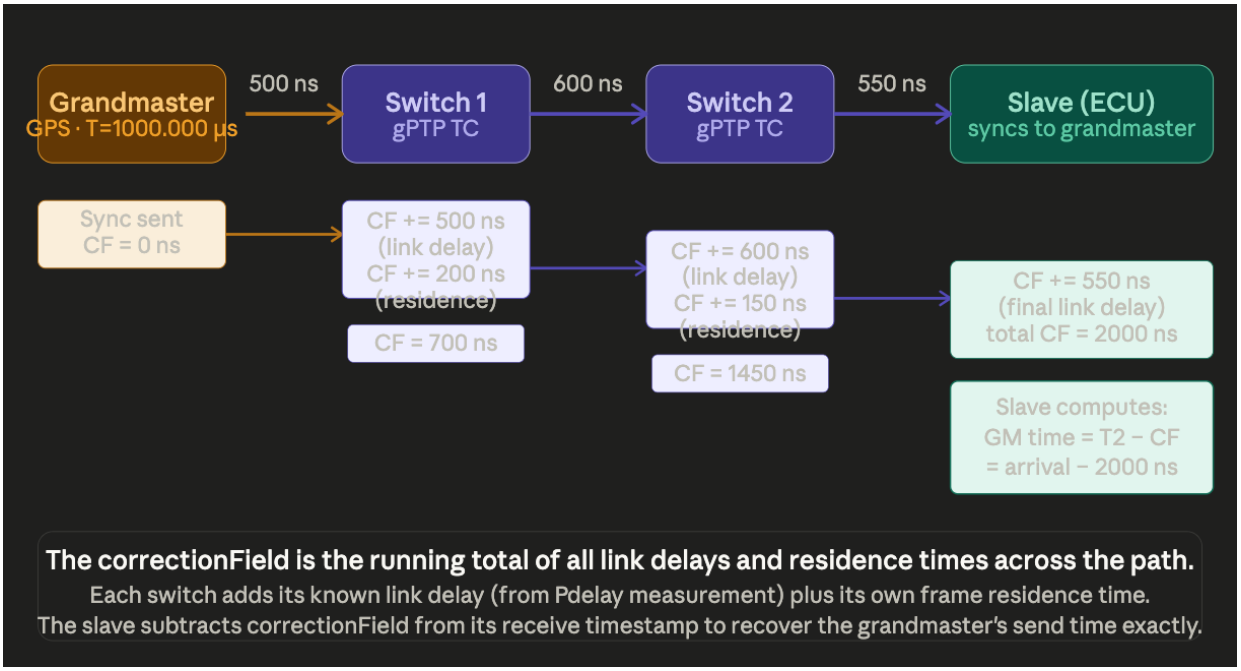
Suppose: T1 = 0 ns, T2 = 520 ns (520 ns forward propagation), T3 = 620 ns (Port B spent 100 ns processing), T4 = 1140 ns (620 ns + 520 ns back).

```
link_delay = ((1140 - 0) - (620 - 520)) / 2 = (1140 - 100) / 2 = 520 ns
```

Port A now knows the one-way delay to its neighbour is 520 ns. It stores this value and uses it when forwarding Sync messages — adding 520 ns to the correctionField so the downstream slave can correct for it. This measurement runs at a configurable interval (typically every 1 second) on every port in the network, simultaneously and independently.

9. correctionField Propagation Through a Multi-Hop Network

The correctionField is a 64-bit field in every PTP Sync and Follow_Up message that accumulates the total of all link delays and switch residence times along the path from grandmaster to slave. By the time the Sync reaches the final slave, its correctionField contains the complete accounting of all latency introduced between the grandmaster and the slave.



Step-by-step example: three hops

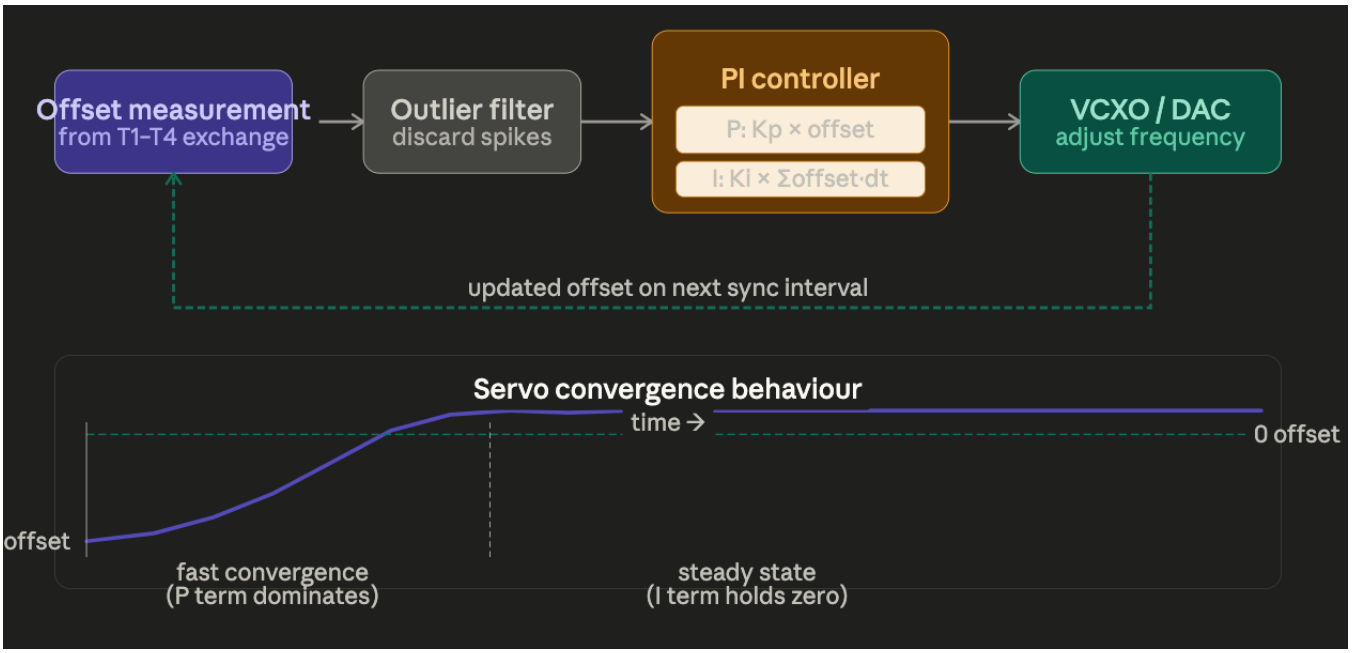
Network element	Action on correctionField	correctionField value after action
Grandmaster	Sends Sync with correctionField = 0	0 ns
Link GM → Switch 1	Switch 1 adds link_delay = 500 ns (measured by Pdelay)	500 ns
Switch 1 (residence)	Switch 1 adds its own frame residence time = 200 ns	700 ns

Network element	Action on correctionField	correctionField value after action
Link SW1 → Switch 2	Switch 2 adds link_delay = 600 ns	1300 ns
Switch 2 (residence)	Switch 2 adds its own frame residence time = 150 ns	1450 ns
Link SW2 → Slave	Slave does not modify correctionField; it just subtracts it	1450 ns (final)
Slave	Computes: GM send time = T2 (receive time) - correctionField = T2 - 1450 ns	Used, not stored

> The slave's clock offset is then: offset = (GM send time) - (local time at T2). The slave drives this offset to zero using its servo. From the slave's perspective, it is directly connected to the grandmaster — all intermediate switch latency has been accounted for.

10. The Servo — How a Slave Adjusts Its Clock

Knowing the offset is not sufficient. The slave must steer its clock to eliminate the offset without causing a discontinuous jump in time. A sudden time jump would corrupt any application using the clock — scheduled transmissions, sensor fusion timestamps, and TSN gate schedules would all become invalid.



The PI controller

PTP clock servos almost universally use a PI (Proportional-Integral) controller. The servo receives a new offset measurement after each Sync/Delay_Req exchange and computes a frequency adjustment for the local oscillator.

Controller term	Formula	Effect
Proportional (P)	output_P = Kp × current_offset	Responds immediately to the current offset. Large offset → large frequency correction. Provides fast initial convergence but leaves a residual steady-state error if used alone.
Integral (I)	output_I = Ki × sum(offset × dt)	Integrates the history of offset measurements. Eliminates steady-state error caused by fixed asymmetries (e.g. a cable that is physically longer in one direction). Converges slowly but completely.
Combined output	freq_adjust = output_P + output_I	Applied to the VCXO (Voltage-Controlled Crystal Oscillator) via a DAC (Digital-to-Analogue Converter). The oscillator frequency is nudged up or down — the clock is steered, never jumped.

Convergence behaviour

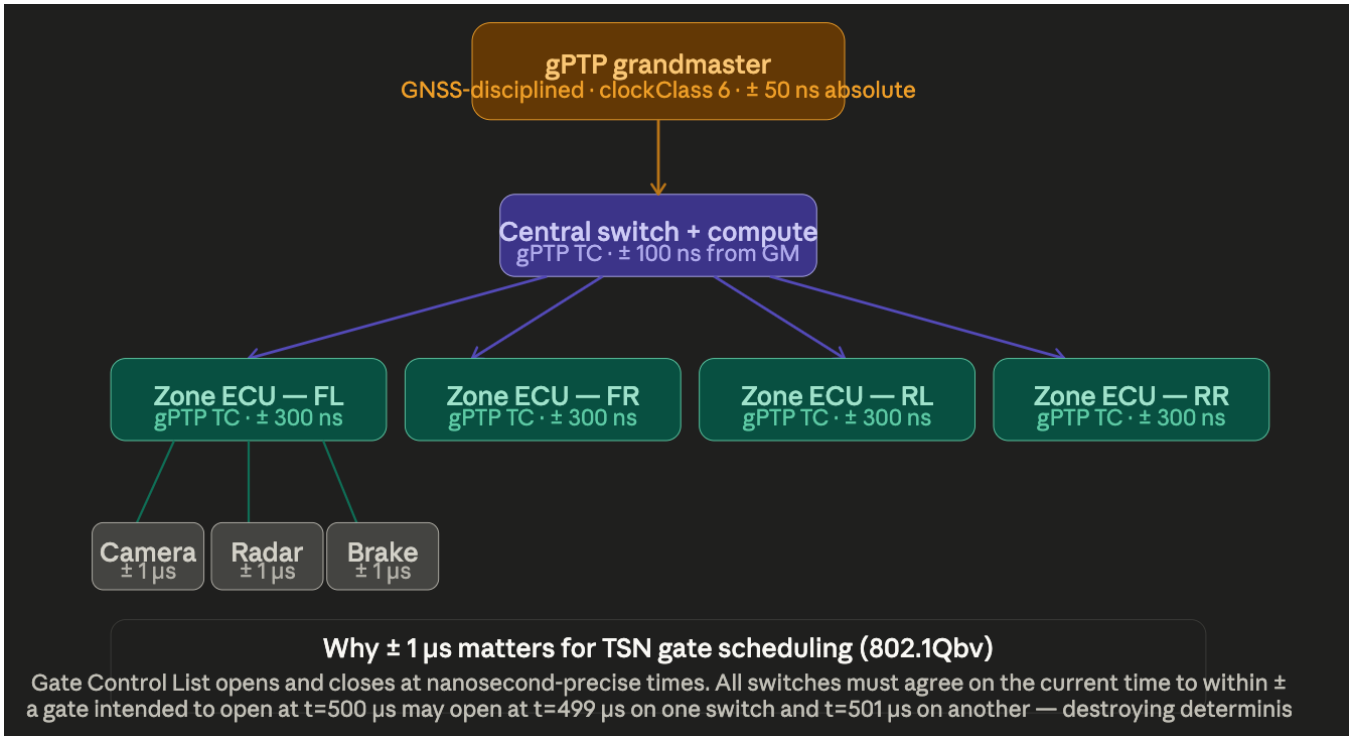
On startup, the offset may be hundreds of microseconds or more. The proportional term drives rapid convergence — within a few sync intervals the offset typically drops to single-digit microseconds. The integral term then takes over to eliminate the remaining bias, reaching steady state in tens to hundreds of seconds depending on the sync interval and PI gains.

In steady state on a well-configured automotive network with hardware timestamping, the residual offset between any two nodes is typically 50–300 nanoseconds. The IEEE 802.1AS-2025 requirement of ± 1 µs across the full vehicle network (up to 8 hops) is comfortably achievable.

> The servo never jumps the clock forward or backward. It only adjusts oscillator frequency (adjFreq, defined in IEEE 1588). If the initial offset is so large that frequency steering alone would take too long, many implementations do a single one-time step adjustment on startup before switching to frequency-only steering.

11. gPTP in Automotive Networks — The Complete Deployment Picture

In a modern zonal vehicle architecture, gPTP provides the common timebase that makes TSN determinism possible. All switches, zone ECUs (Electronic Control Units), and sensor nodes must agree on the current time to within ± 1 µs for the 802.1Qbv Time-Aware Shaper (TAS) gate schedules to function correctly.



Why ± 1 µs matters for TSN gate scheduling

The 802.1Qbv Gate Control List (GCL) opens and closes egress queue gates at precisely scheduled times. Consider a safety-critical stream (PCP = 7, brake-by-wire command) with a gate window that opens at t = 500 µs in every 1 ms cycle. If two switches in the path disagree on the current time by 2 µs, one switch may open its gate at t = 498 µs and the other at t = 502 µs. The frame arrives at the first switch's open gate but the second switch's gate has already closed — the frame is delayed by a full cycle (1 ms). For brake-by-wire, this is a functional safety violation under ISO 26262.

Typical accuracy budget — four-hop vehicle network

Network level	Accuracy relative to GPS grandmaster	Clock type
GPS grandmaster (GNSS receiver)	± 50 ns (traceable to UTC)	Grandmaster (OC master)
Central compute switch — hop 1	± 100 ns	gPTP Transparent Clock
Zone ECU switch — hop 2	± 300 ns	gPTP Transparent Clock
ECU sensor/actuator — hop 3	± 600 ns	OC slave

Network level	Accuracy relative to GPS grandmaster	Clock type
Worst case across 8 hops	< ± 1 µs	IEEE 802.1AS requirement

Acronym reference — all terms used in this document

Acronym	Full name	Brief definition
PTP	Precision Time Protocol	IEEE 1588 — sub-microsecond clock synchronisation for packet networks
gPTP	Generalised Precision Time Protocol	IEEE 802.1AS — PTP profile for bridged LANs and TSN
GM	Grandmaster clock	Single authoritative time source per PTP domain; typically GPS-disciplined
OC	Ordinary Clock	Single-port PTP device; either pure master or pure slave endpoint
BC	Boundary Clock	Multi-port device: slave upstream, master downstream; terminates PTP per hop
TC	Transparent Clock	Switch that passes Sync through and adds residence time to correctionField
BMCA	Best Master Clock Algorithm	Distributed election protocol for grandmaster selection
SFD	Start Frame Delimiter	Byte 0xAB marking start of Ethernet frame; hardware timestamp is captured at SFD
MDI	Media Dependent Interface	The physical wire boundary of a network port; where hardware timestamps are taken
VCXO	Voltage-Controlled Crystal Oscillator	Oscillator whose frequency the servo adjusts via DAC voltage
DAC	Digital-to-Analogue Converter	Converts servo PI controller output to analogue voltage driving the VCXO
CDR	Clock and Data Recovery	PHY PLL that extracts a clock from incoming data transitions
MDIO	Management Data Input/Output	Two-wire serial interface for reading/writing PHY and PTP registers
CF	correctionField	64-bit PTP message field accumulating all link delays and residence times on the path
TAS	Time-Aware Shaper	IEEE 802.1Qbv gate scheduling mechanism; requires gPTP for timing
GCL	Gate Control List	The schedule table that controls when each TSN queue gate opens and closes
NTP	Network Time Protocol	RFC 5905 — millisecond-accuracy time sync; insufficient for TSN applications
PI	Proportional-Integral (controller)	Servo algorithm: P term gives fast response, I term eliminates steady-state error

Acronym	Full name	Brief definition
TSN	Time-Sensitive Networking	IEEE 802.1 suite of standards for deterministic Ethernet, built on gPTP
GNSS	Global Navigation Satellite System	GPS, Galileo, GLONASS etc. — source of traceable UTC time for PTP grandmasters
ECU	Electronic Control Unit	Automotive embedded computer; typically an OC slave in a gPTP network
UTC	Coordinated Universal Time	The international civil time standard; PTP grandmasters are traceable to UTC

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